

Raymond Xu

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EDUCATION

Northeastern University

Boston, MA

Bachelor of Science in Computer Engineering & Computer Science

Sep. 2024 – May 2028

- **GPA: 3.95/4.0**, Dean's List, Eta Kappa Nu, Tau Beta Pi, Rev Cohort 4
- Coursework: Algorithms & Data, Computer Systems, Object-Oriented Design, Embedded Design: Robotics

EXPERIENCE

Software Engineer Intern

Jun. 2026 – Present

NASA Jet Propulsion Laboratory

La Cañada Flintridge, CA

- Integrated autonomous, orbital-imagery-driven path-planner into rover traverse planning tool, replacing manual waypoint routing with AI planned drive routes, by bridging Python CLI planner to React/Three.js frontend.
- Built Python CLI tool automating manual translation of human-made activity plans for Mars Perseverance rover into machine readable command sequences via an AI/LLM LangChain pipeline trained on 800+ mission plans.
- Resolved 5 issues on production rover traverse planning tool, boosting performance and fixing 3D-rendering bugs.

Software Engineer Intern

Jan. 2026 – May 2026

NASA Kennedy Space Center

Merritt Island, FL

- Modernized runtime config retrieval for Artemis III/IV missions by engineering 52+ C++ ActiveMQ request & response message classes, replacing legacy config files with centralized database under Launch Control System.
- Ensured software correctness by authoring CxxTest unit test suites achieving 90%+ coverage across all Master Console Operator Messaging classes and shared System Monitoring & Control data types.
- Enhanced operator troubleshooting across all 4 firing rooms by extending a Java/FXML GUI giving a holistic view of individual servers and workstations, adding CPU, RAM, and disk usage and mount location visualizations.

Software Engineer Intern

Jul. 2025 – Aug. 2025

Mosaic (YC W25)

San Francisco, CA

- Optimized agentic video editing pipeline by 40% by redesigning a React Flow-based node editor with pausing, selective restarts, and internal versioning for real-time debugging.
- Engineered a Remotion-based video editor in TypeScript/Next.js with synchronized timeline and inspector, boosting render performance by 80% and enabling multimodal video agent iteration.
- Built Python/FastAPI endpoints to manage Supabase data and orchestrate Gemini API calls for agent reasoning.

AI/ML Fellow

Apr. 2025 – Jan. 2026

Break Through Tech

Cambridge, MA

- Selected from 3000+ applicants for 12-month program at MIT including ML coursework with Cornell faculty.
- Collaborated with MathWorks on Human Motion Recognition project, training Random Forest models for activity, walking speed, sleep-wake classification on IMU HARTH dataset, achieving 94% accuracy.
- Built real-time activity classifier (React/FastAPI) processing accelerometer data streamed from MATLAB Mobile.

PROJECTS

LLM Auto-Formalization for Verified Lean Proofs | *Python, PyTorch, vLLM, LoRA*

May 2026 – Present

- Fine-tuned 8B-parameter LLMs with LoRA to generate formal Lean proofs verified by automated theorem prover.
- Built pass@k evaluation pipeline for DeepSeek-Prover via vLLM with 8 parallel Lean REPL verification workers.
- Enforced proof correctness via sandboxed lake compiler and axiom-checking comparator to reject unsound proofs.

Voice Controlled Robotic Hand Brace | *C++, Flutter, Arduino, Bluetooth, SolidWorks*

Jan. 2025 – April 2025

- Developed C++ firmware on Arduino controlling 5 independent servo-actuated fingers with real time positioning.
- Designed a Bluetooth communication protocol linking a mobile app to the microcontroller, supporting voice commands and 20+ customizable grip presets for stroke and arthritis patients.

TECHNICAL SKILLS

Languages: Java, Python, C/C++, C#, JavaScript, TypeScript, HTML/CSS, SQL, Swift, Matlab, Lua

Frameworks: React, Next.js, Tailwind, Zustand, Three.js, Node.js, Express, FastAPI, Spring Boot, JUnit, PyTorch

Tools: Git, GitHub Actions, Jenkins, Bash, Linux, Cursor, Claude Code, Figma, Unity, Godot, SolidWorks, Blender

Databases: PostgreSQL, MySQL, SQLite, Supabase, MongoDB, Redis

Cloud/Infra: Docker, Kubernetes, AWS (S3, EC2, Lambda), GCP, Vercel